A TOY MODEL OF CONTINUOUS MEASUREMENT

Why the hell is my plot so awful ?

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Département de Physique
École Normale Supérieure

DISCLAIMER

Work done in collaboration with **Denis Bernard** and **Michel Bauer**.
The idea of the toy model was found with **Irénée Frérot**.

Small part of a larger work in progress: not even on the arxiv yet

Mathematical curiosity & interesting physical effect ? Glad to have your point of view at the end.

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WHY

- Measurement is a subtle conceptual issue in quantum mechanics
- There are interesting models of continuous measurement in quantum mechanics
- Need for a simple classical counterpart to see what is genuinely quantum

OUTLINE

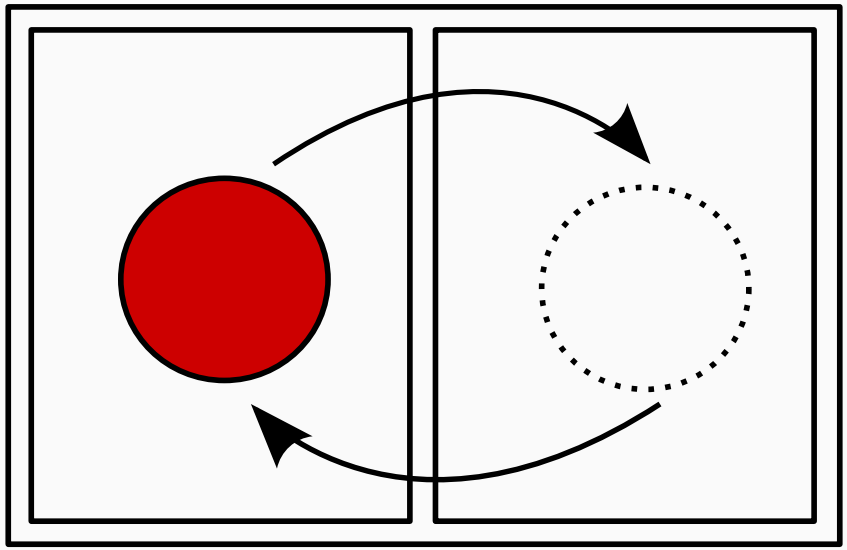
A simple classical model

In the quantum realm

A SIMPLE CLASSICAL MODEL

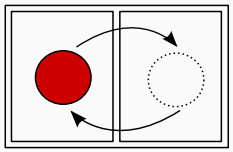
Repeated fuzzy measurements

Consider a classical 2-state Markov process. Typically a particle randomly jumping between two compartments of a box.

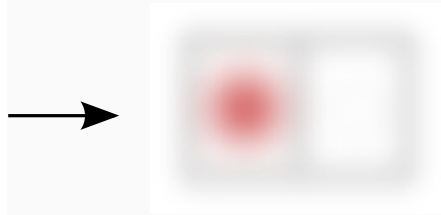


Repeated fuzzy measurements

The box is very small and we take fuzzy pictures from far away.







Every picture gives a tiny bit of information about where the particle is.

Mathematical Model

- The particle jump rate is λ, the number of jumps is a Poisson process.

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$$\mathbb{P}(\delta_n = 1 | \text{particle on the left}) = \frac{1+\epsilon}{2}$$

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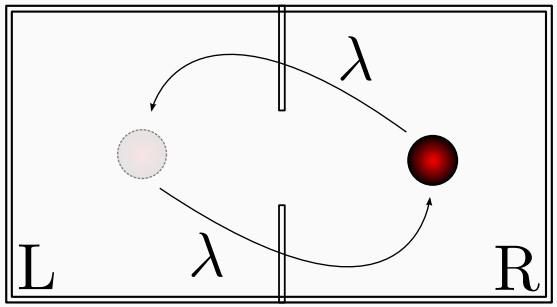
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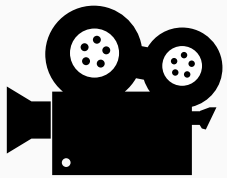
$$\mathbb{P}(\delta_n = 1 | \text{particle on the right}) = \frac{1-\epsilon}{2}$$

- We are interested in:

$$Q_n = \mathbb{P}(\text{particle on the left at time n} | \text{all the pictures before n})$$

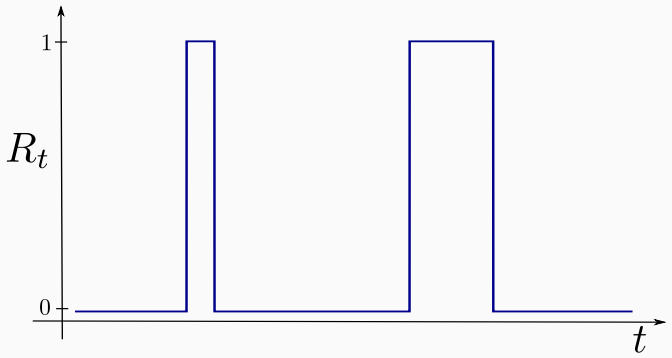
Summary

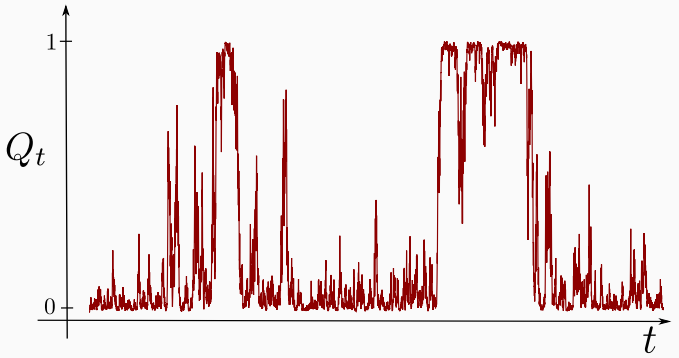






$\delta_t = \ldots \qquad \delta_3 = +1 \qquad \delta_2 = +1 \qquad \delta_1 = -1$





Mathematical Model

To compute Q_{n+1} knowing Q_n we need to:

- Incorporate the measurement result δ_n using **Bayes rule**:

$$\begin{aligned} Q_{n+1} &= \mathbb{P}(\text{left at n+1} | Q_n \& \delta_{n+1}) \\ &= \frac{\mathbb{P}(\delta_{n+1} | \text{left at n+1})\, \mathbb{P}(\text{left at n+1} | Q_n)}{\mathbb{P}(\delta_{n+1} | Q_n)} \end{aligned}$$

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- Incorporate the fact that **we know that the particle tends to jump** during the time interval:

$$\mathbb{P}(\text{left at n+1} | Q_n) = (1 - \lambda) Q_n + \lambda (1 - Q_n)$$

Mathematical Model

At the continuous limit, for extremely fuzzy pictures (and a proper rescaling) we have:

$$dQ_t = \lambda \left(\frac{1}{2} - Q_t \right) dt + \gamma \, Q_t(1 - Q_t) \times \text{white noise} \tag{1}$$

with γ^2 the rate at which pictures are taken.

Mathematical Model

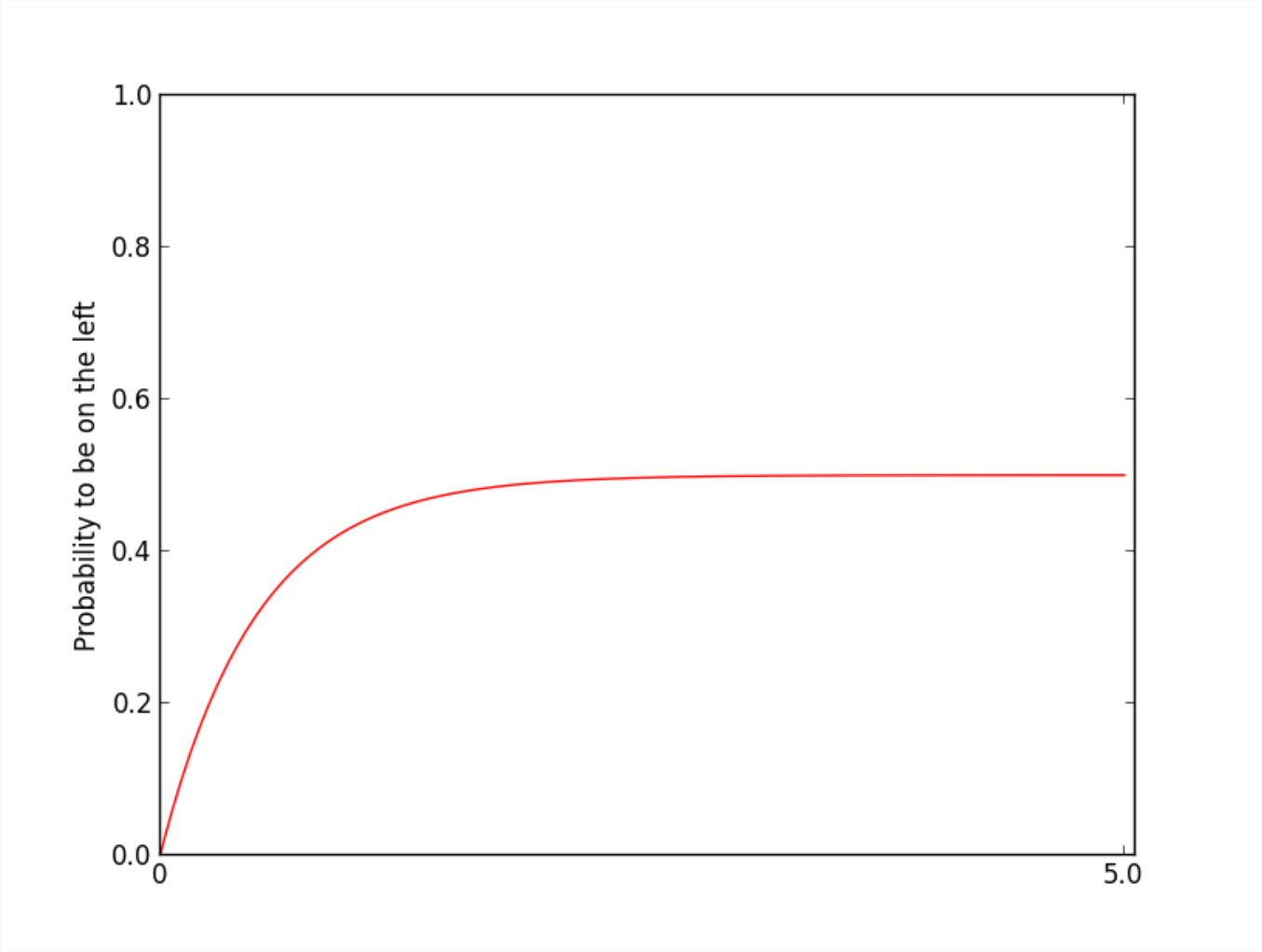
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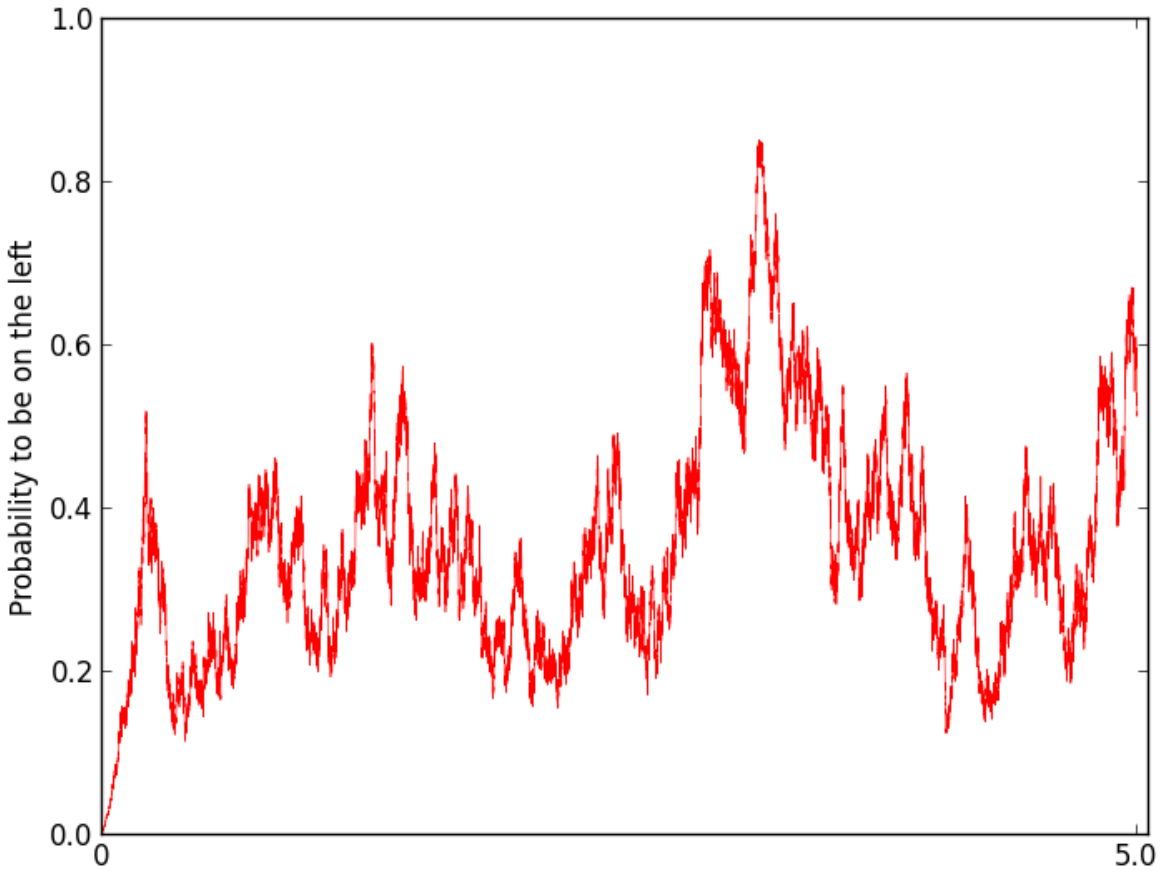
The continuous limit is only needed for the closed form results, all the rest is true in the discrete case. **What follows is not an artifact of the diffusive limit.**

Results



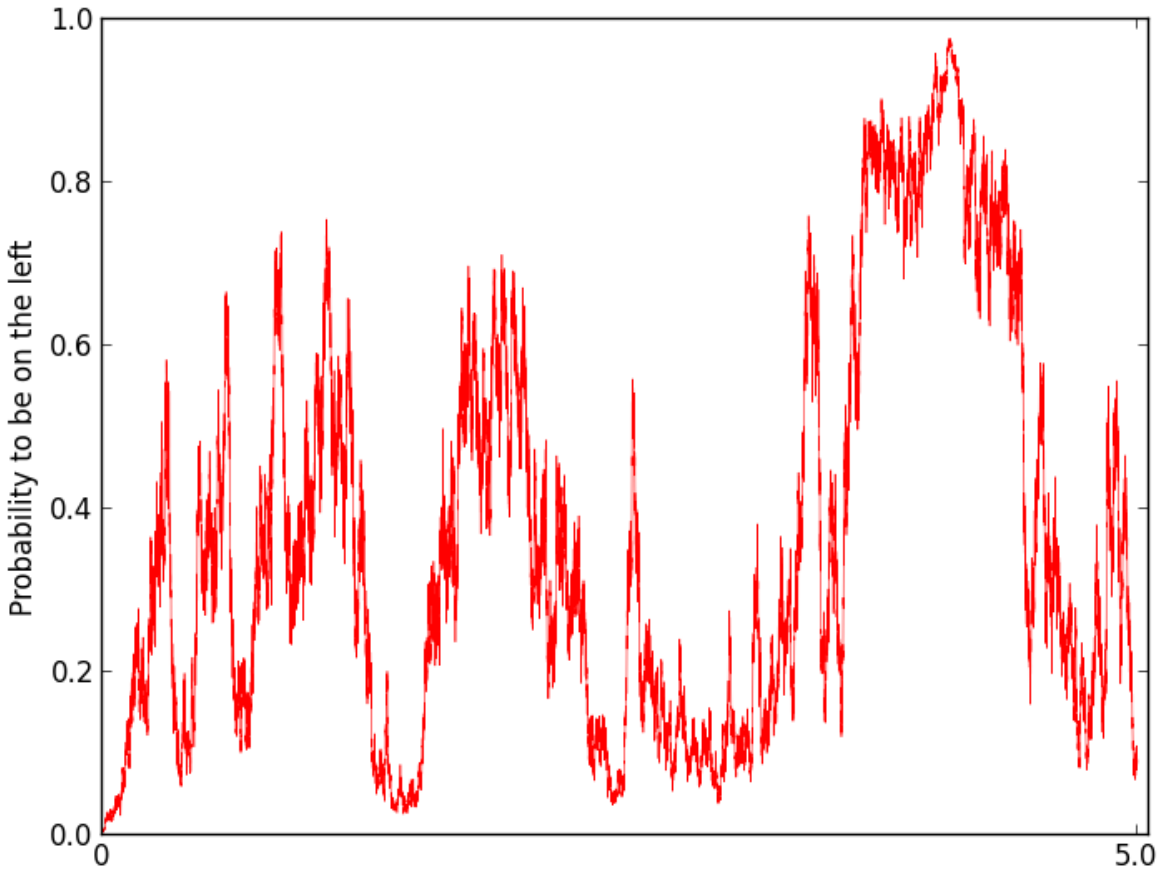
No Measurements

Results



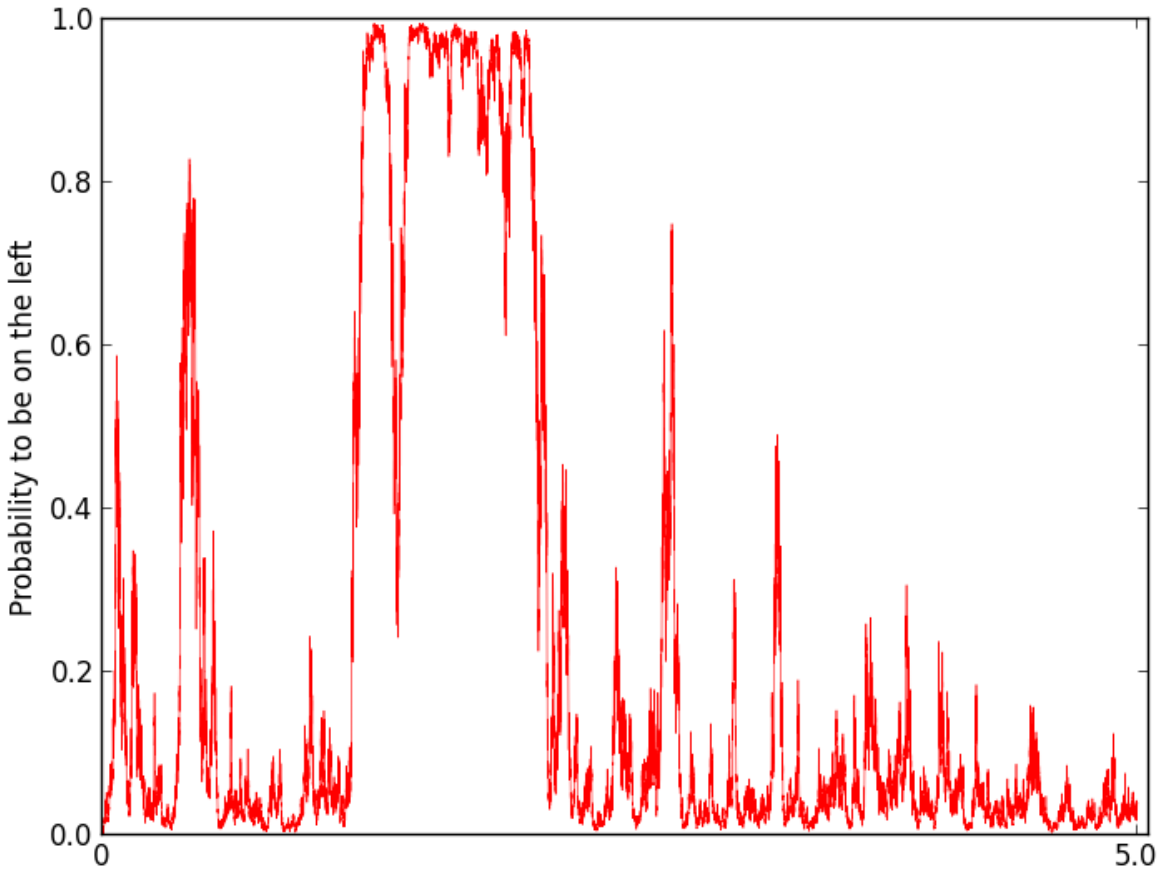
$\gamma = 0.5$

Results



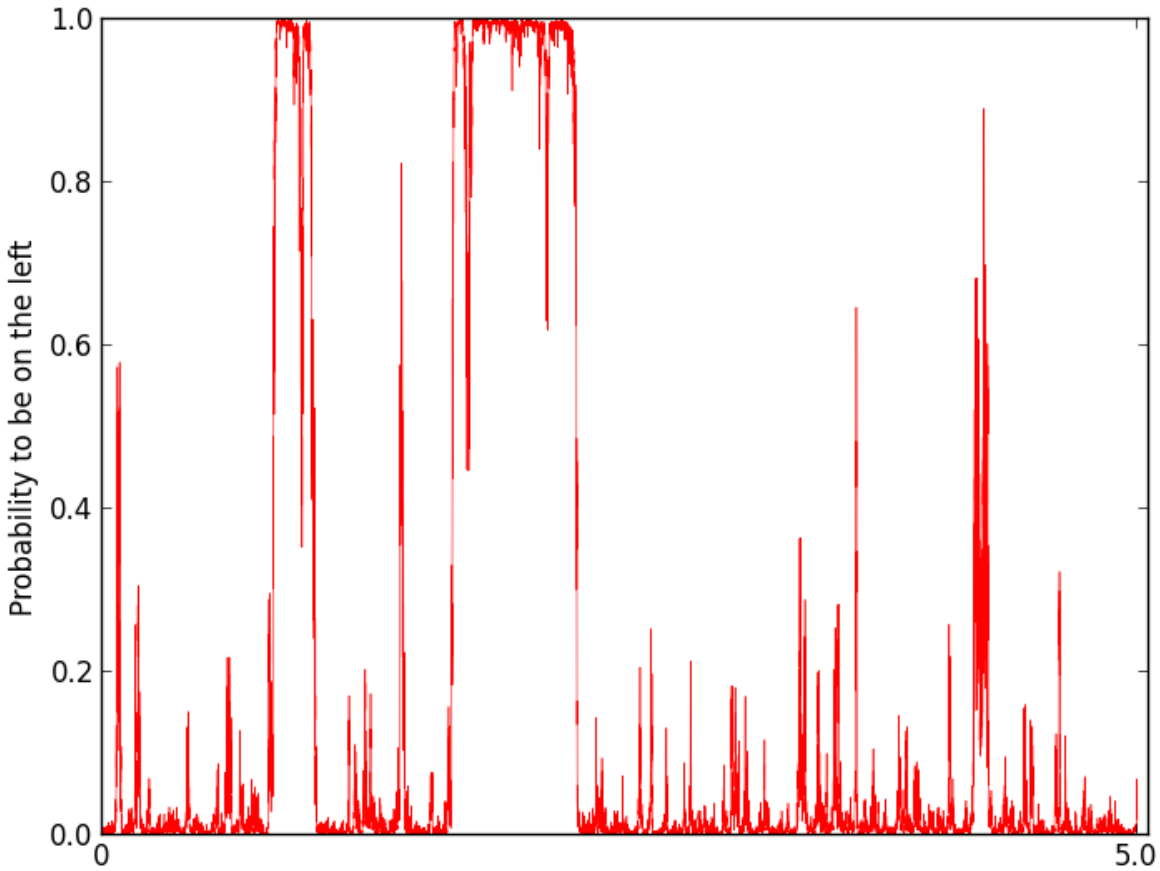
$\gamma = 1.0$

Results



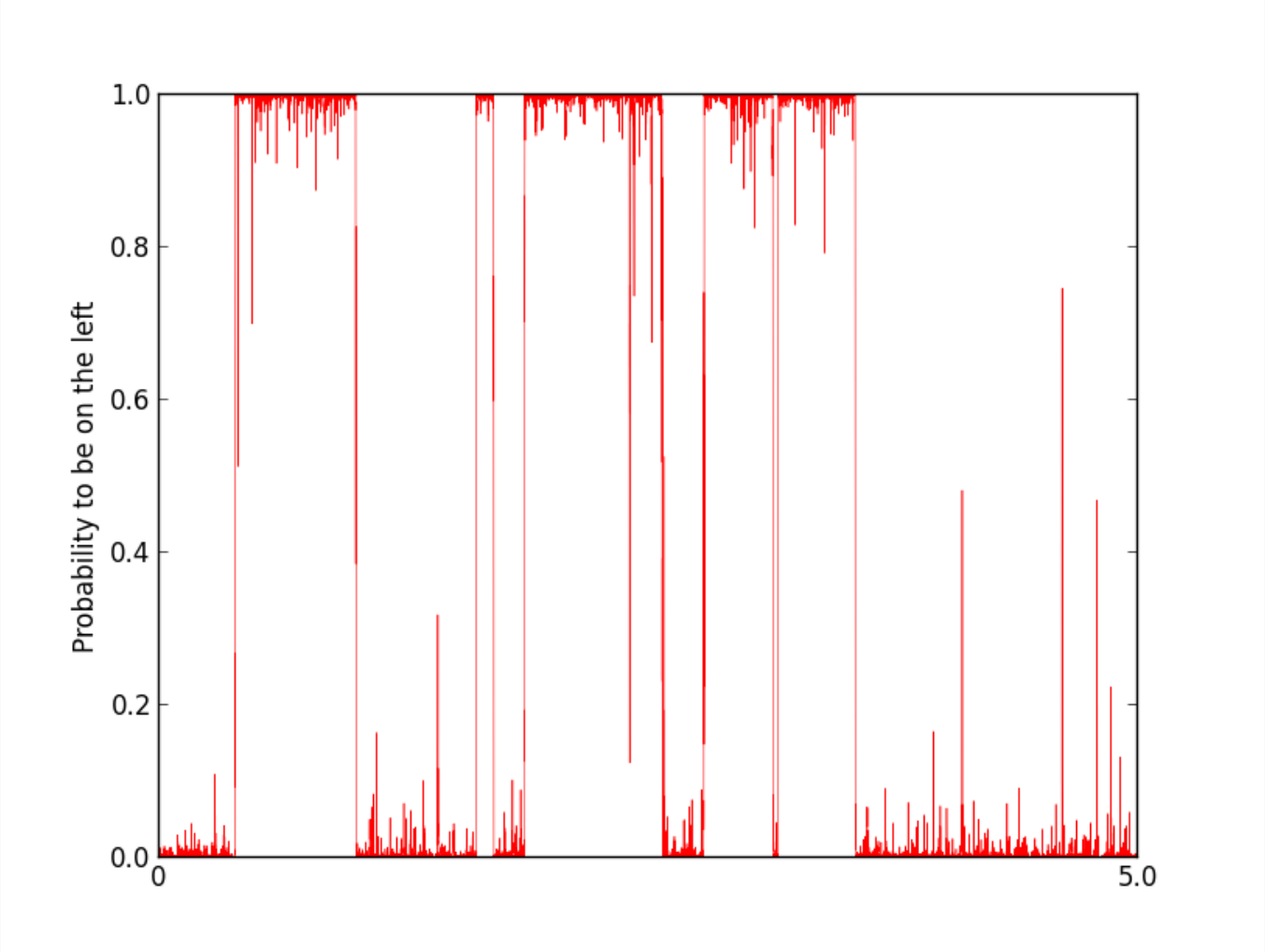
$\gamma = 2.0$

Results



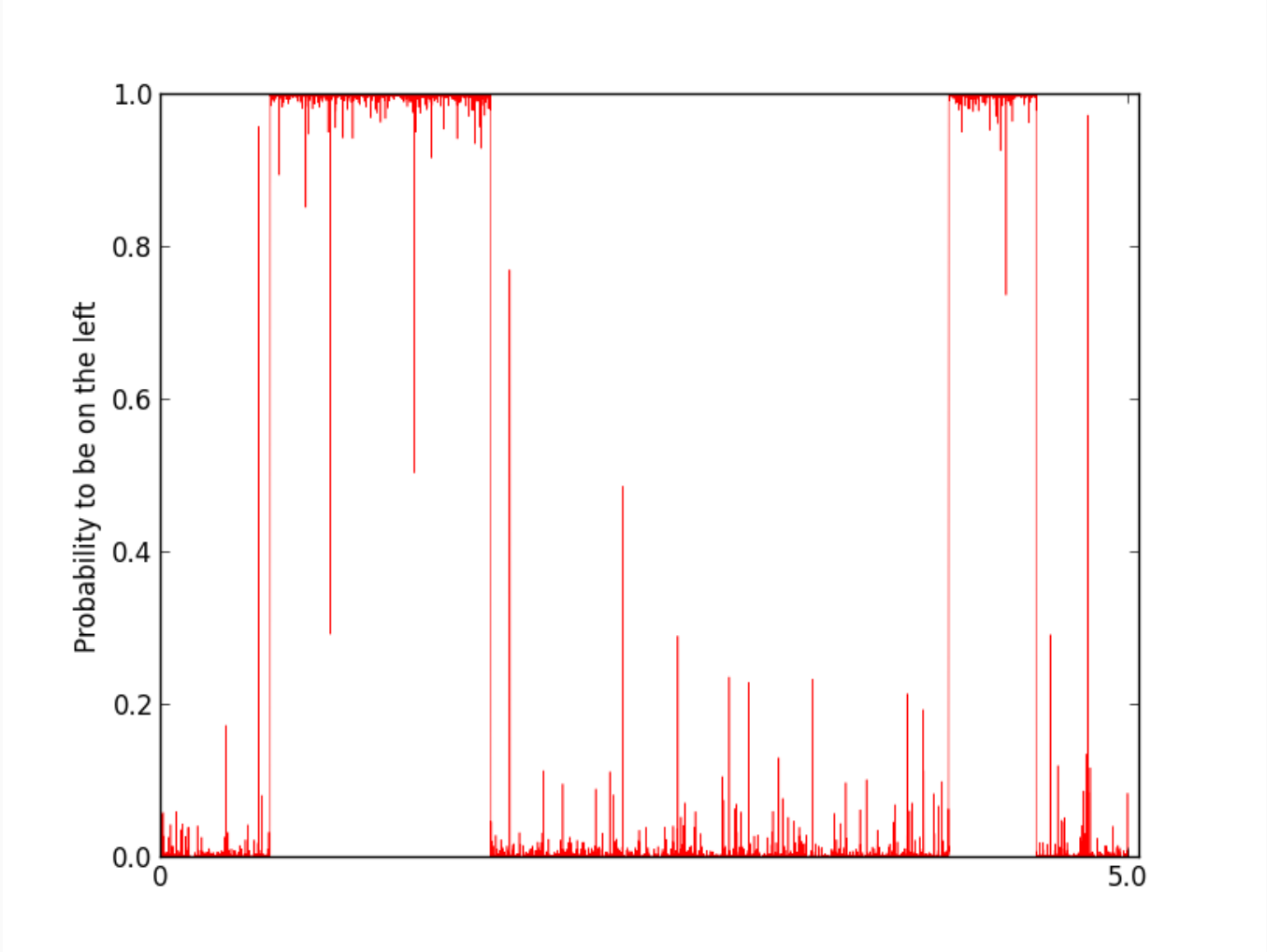
$\gamma = 5.0$

Results



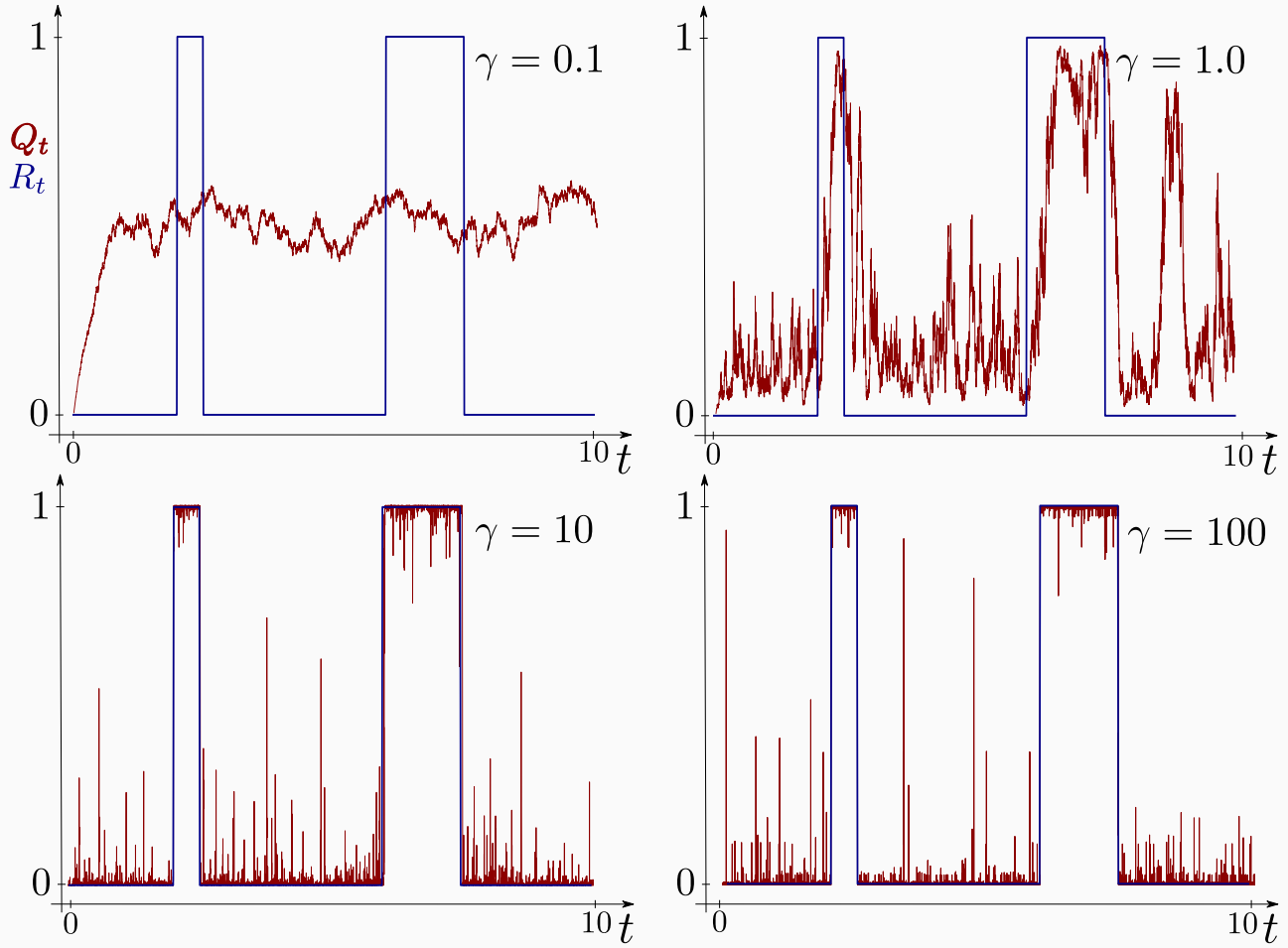
$\gamma = 20.0$

Results

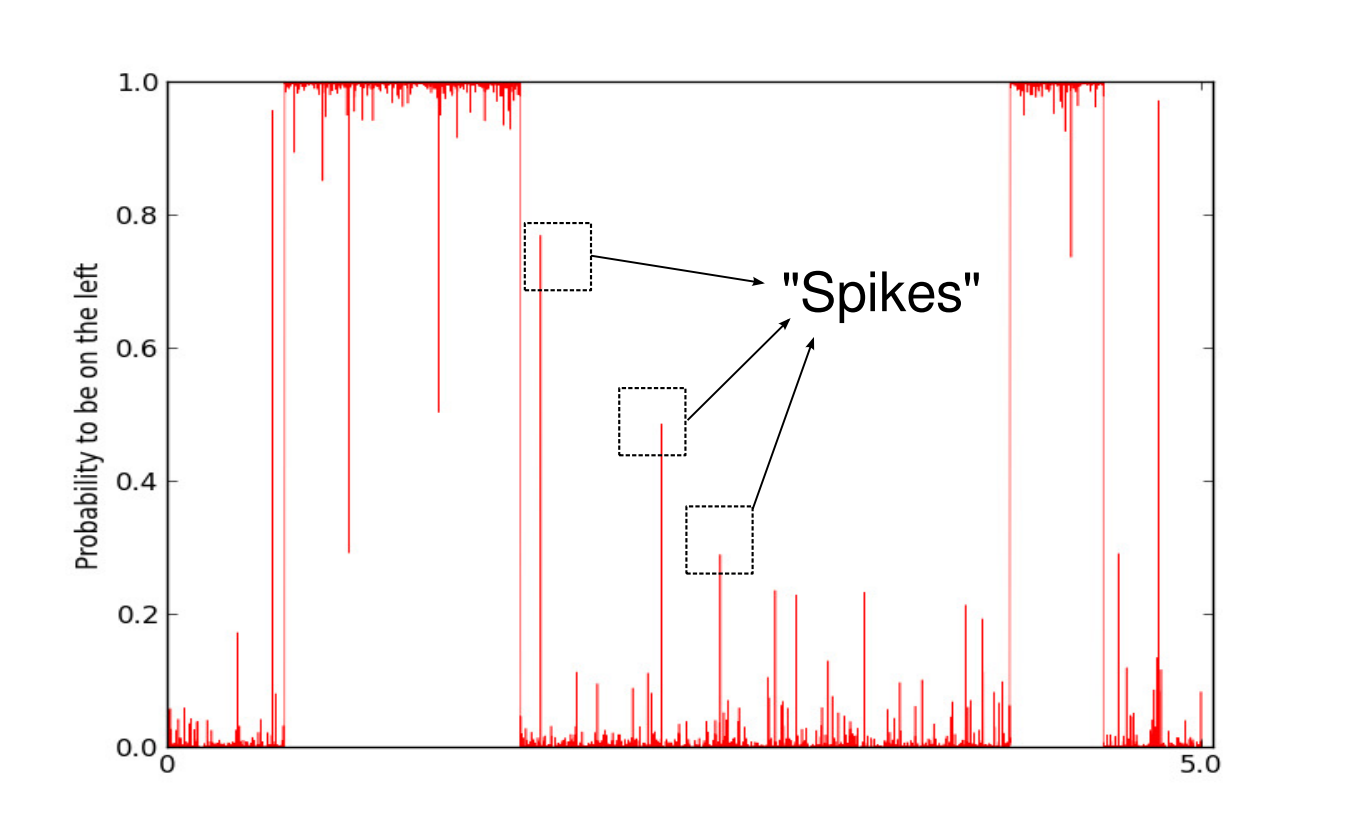


$\gamma = 100.0$, no difference with $\gamma = +\infty$

Summary


1
0
Q_t
R_t
$\gamma = 0.1$
0
10
t
1
0
$\gamma = 1.0$
0
10
t
1
0
$\gamma = 10$
0
10
t
1
0
$\gamma = 100$
0
10
t

What we call spikes:



Spikes do not disappear when $\gamma \to +\infty$! They just become sharper and sharper.

Theorem

When $\gamma \to \infty$, on a time interval without jumps, the process giving the top of the spikes $\tilde{Q}_t$ is a Poisson process of intensity:

$$d\nu = \lambda dt \, \frac{dQ}{Q^2}$$

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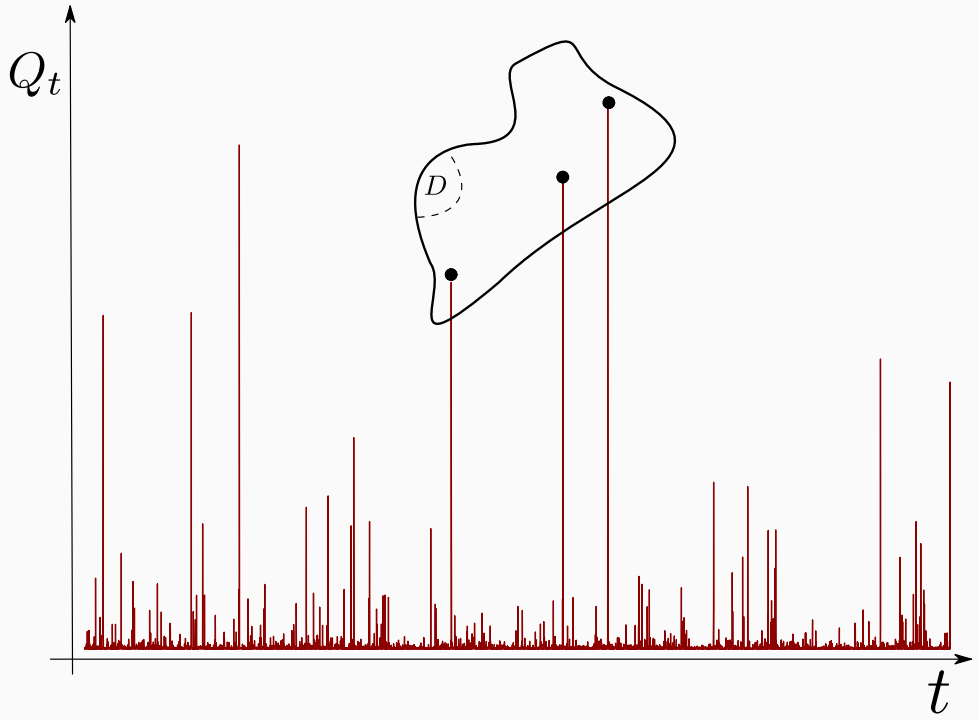
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- Infinitely many small spikes on any time interval
- Scale invariant

General Poisson Process



The number N of spikes in the domain D is a standard Poisson process of intensity μ, i.e. $P(N) = \frac{\mu^N}{N!} e^{-\mu}$ with:

$$\mu = \int_D d\nu$$

Classical meaning of spikes

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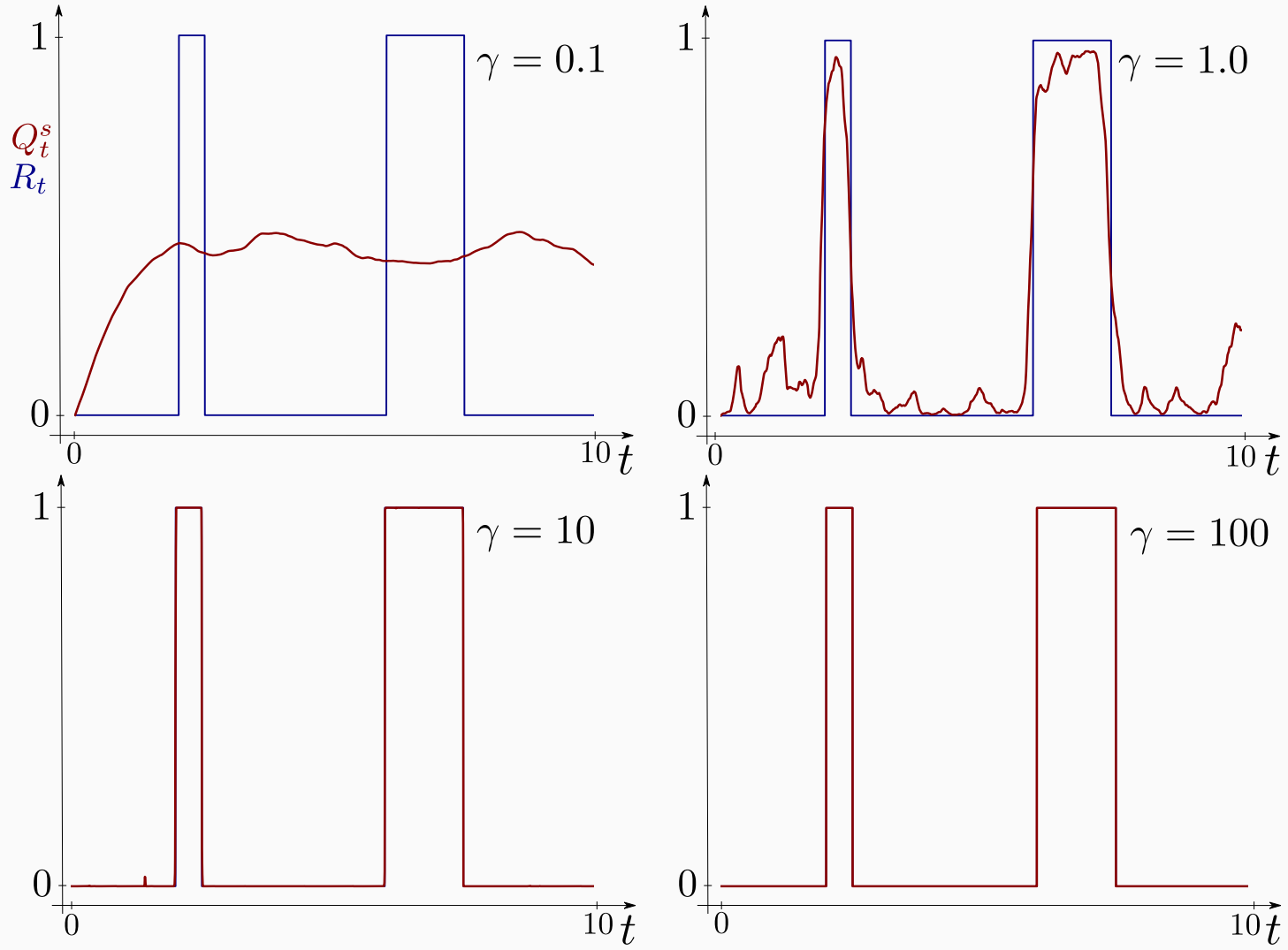
- Spikes can be seen as a **artifact** of Bayesian inference, nothing real is intrinsically ”spiky”.

Classical meaning of spikes

The implications of spikes in this classical model are benign:

- Spikes can be seen as a **artifact** of Bayesian inference, nothing real is intrinsically ”spiky”.
- Spikes can be removed by forward-backward estimation (smoothing) or more brutally by low-pass filtering.

Forward-backward estimation



IN THE QUANTUM REALM

A quantum system with spikes

Consider a two level system (a qubit) with Hamiltonian $H = \frac{\omega}{2}\sigma_x$ with σ_z continuously monitored at a rate γ.

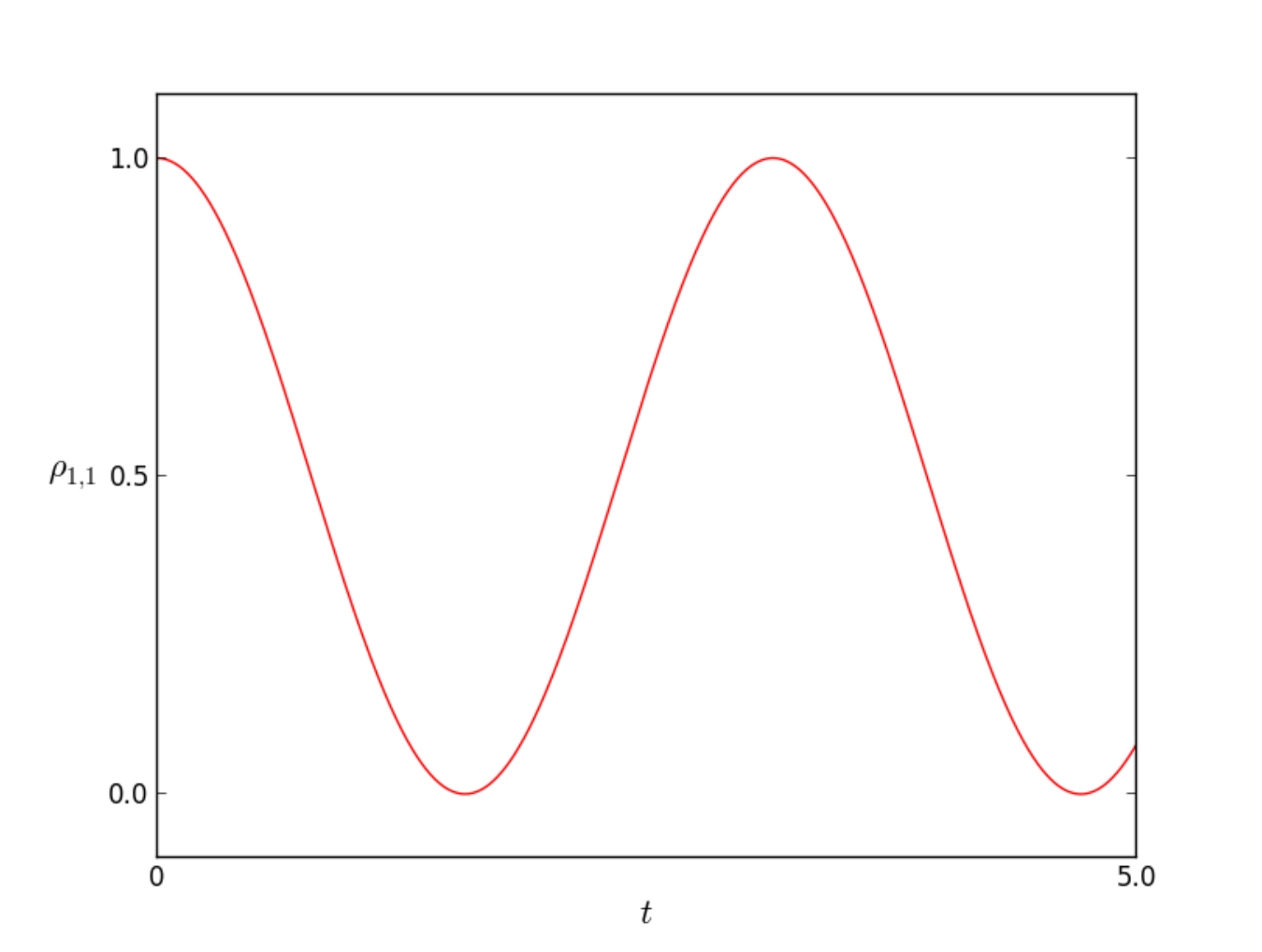
A quantum system with spikes

Consider a two level system (a qubit) with Hamiltonian $H = \frac{\omega}{2}\sigma_x$ with σ_z continuously monitored at a rate γ.

The evolution is given by the stochastic master equation:

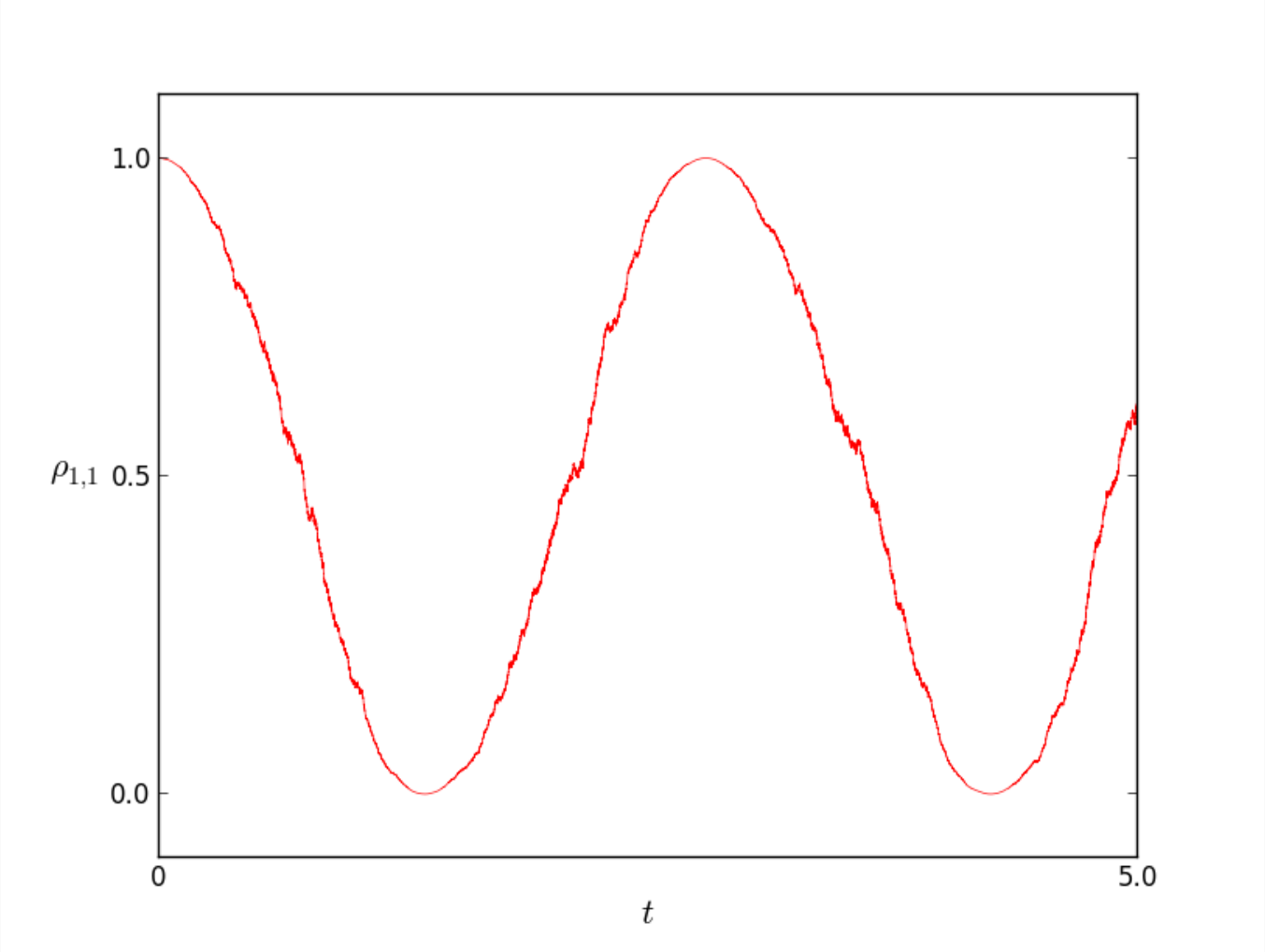
$$d\rho_t = -i\frac{\omega}{2}[\sigma_x, \rho_t]dt - \frac{\gamma^2}{2}\left[\sigma_z\left[\sigma_z, \rho_t\right]\right]dt + \gamma\left(\sigma_z\rho_t + \rho_t\sigma_z - 2tr\sigma_z\rho_t\right)dW_t$$

Results



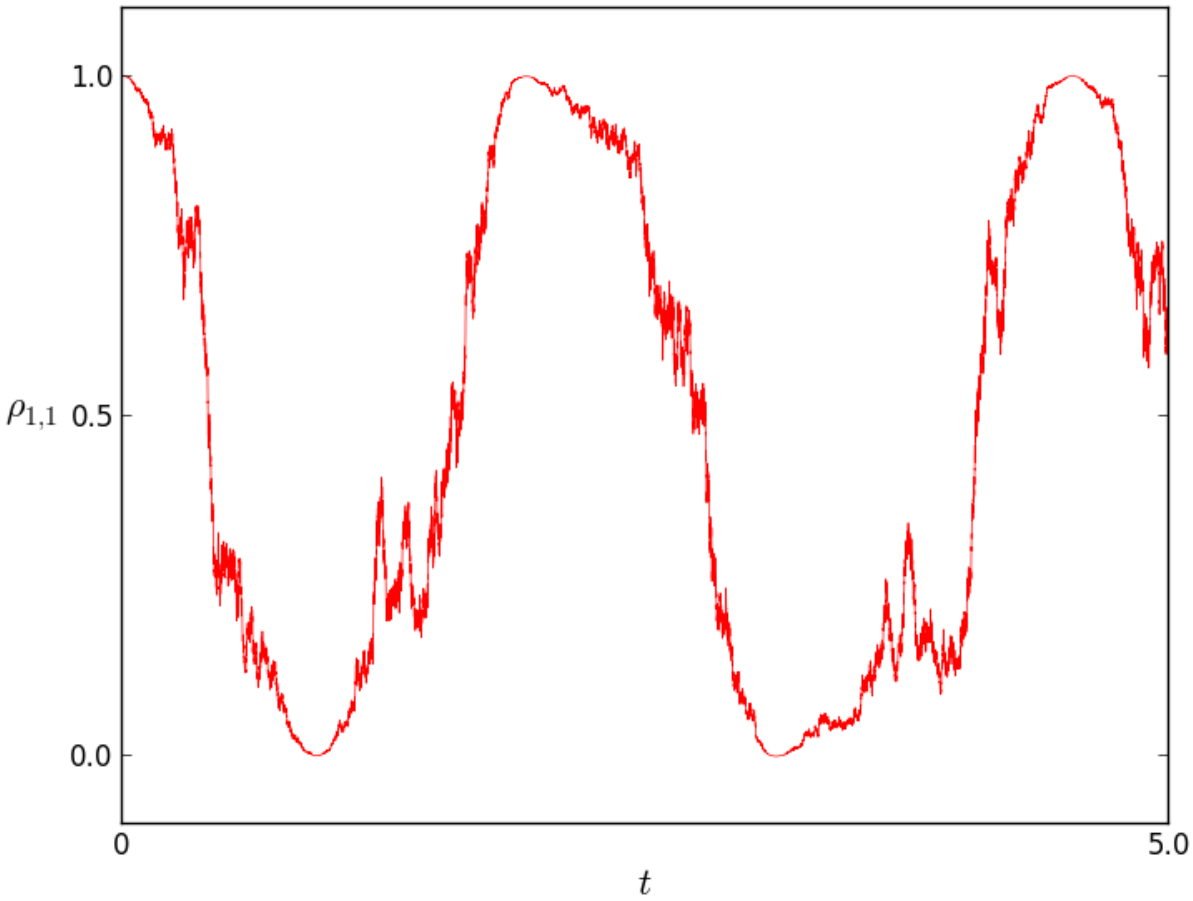
Without measurement $\gamma = 0.0$

Results



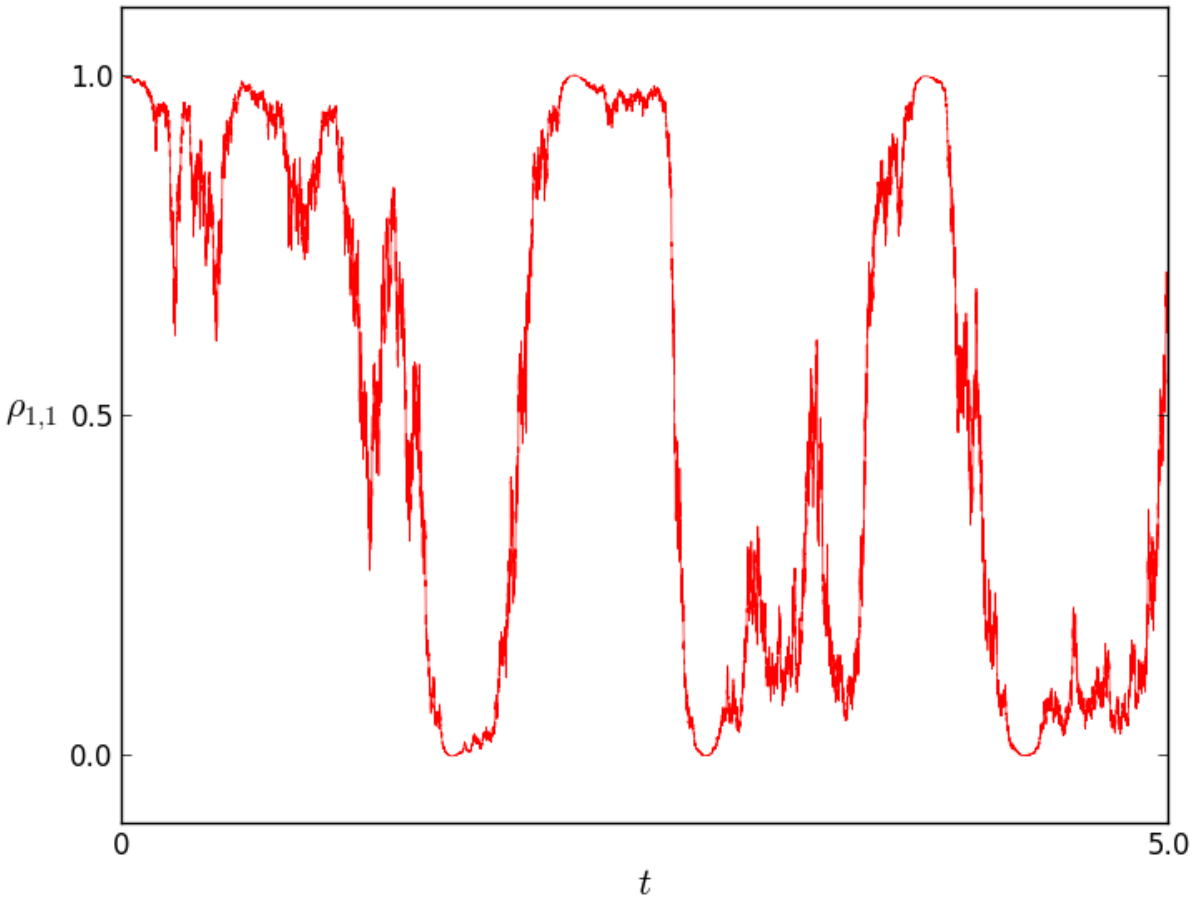
$\gamma = 0.1$

Results



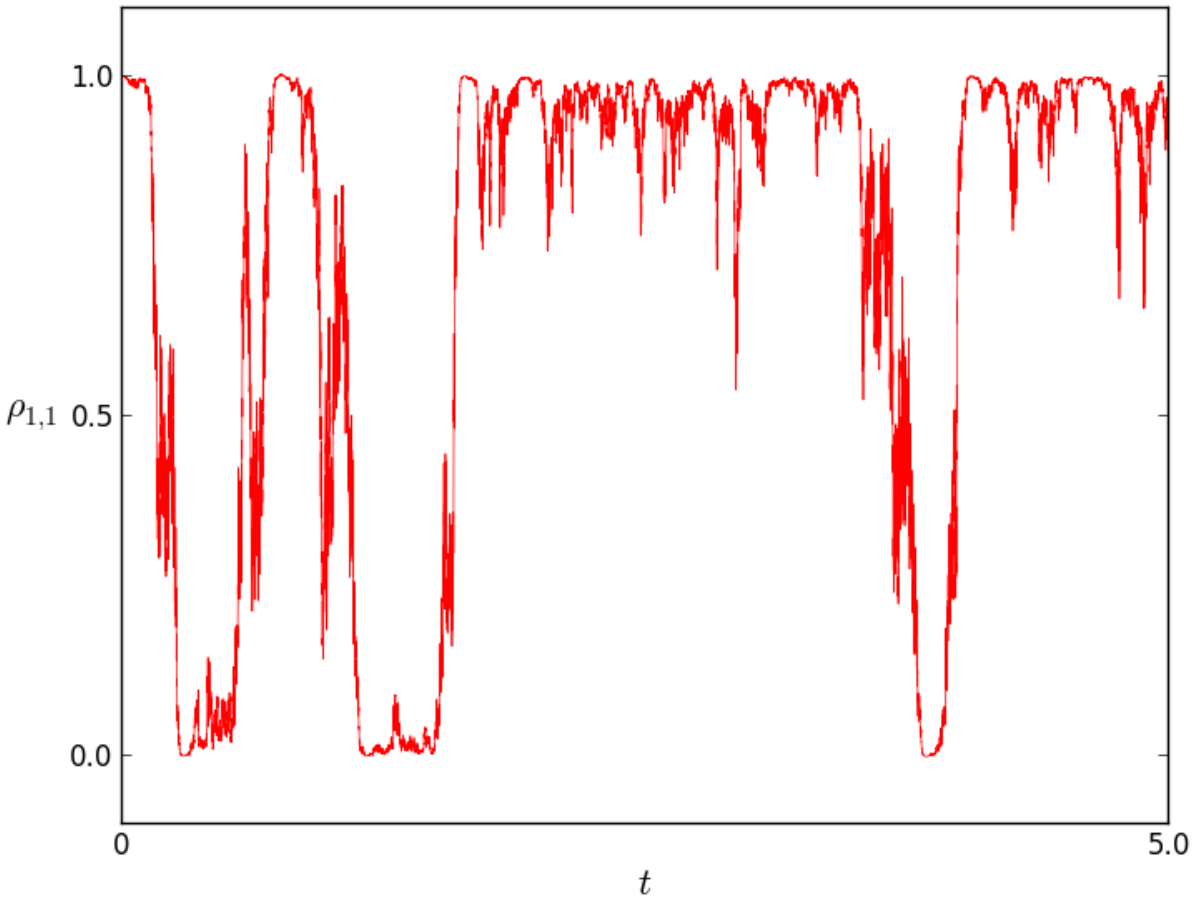
$\gamma = 0.5$

Results



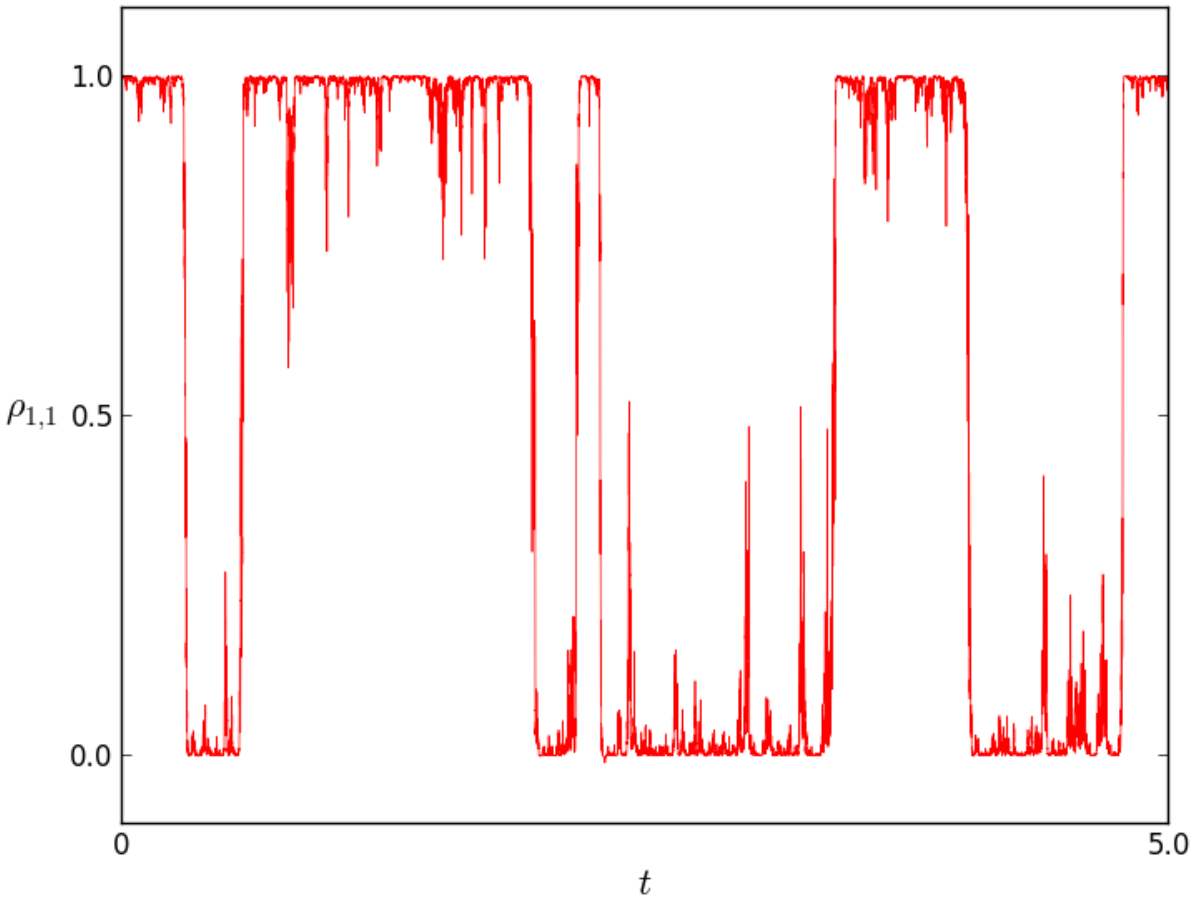
$\gamma = 1.0$

Results



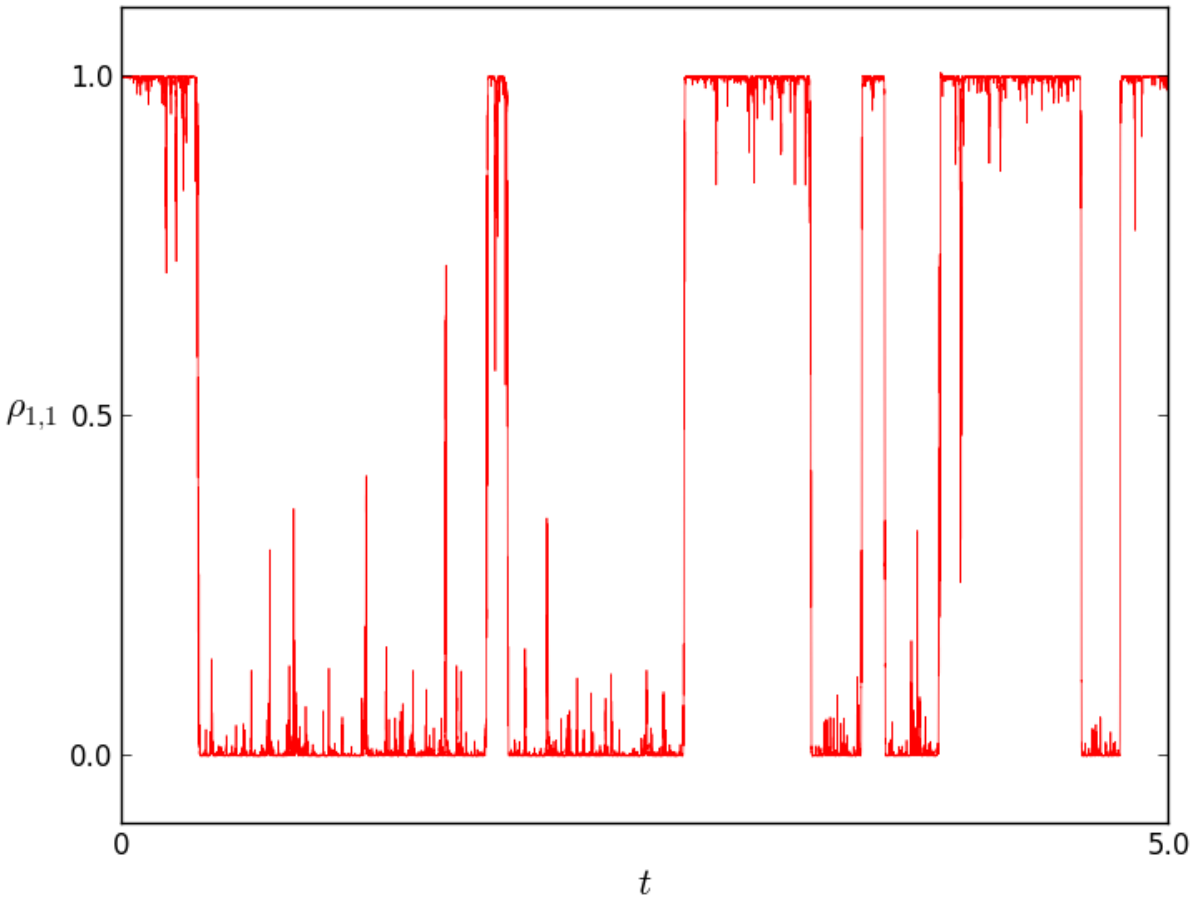
$\gamma = 2.0$

Results



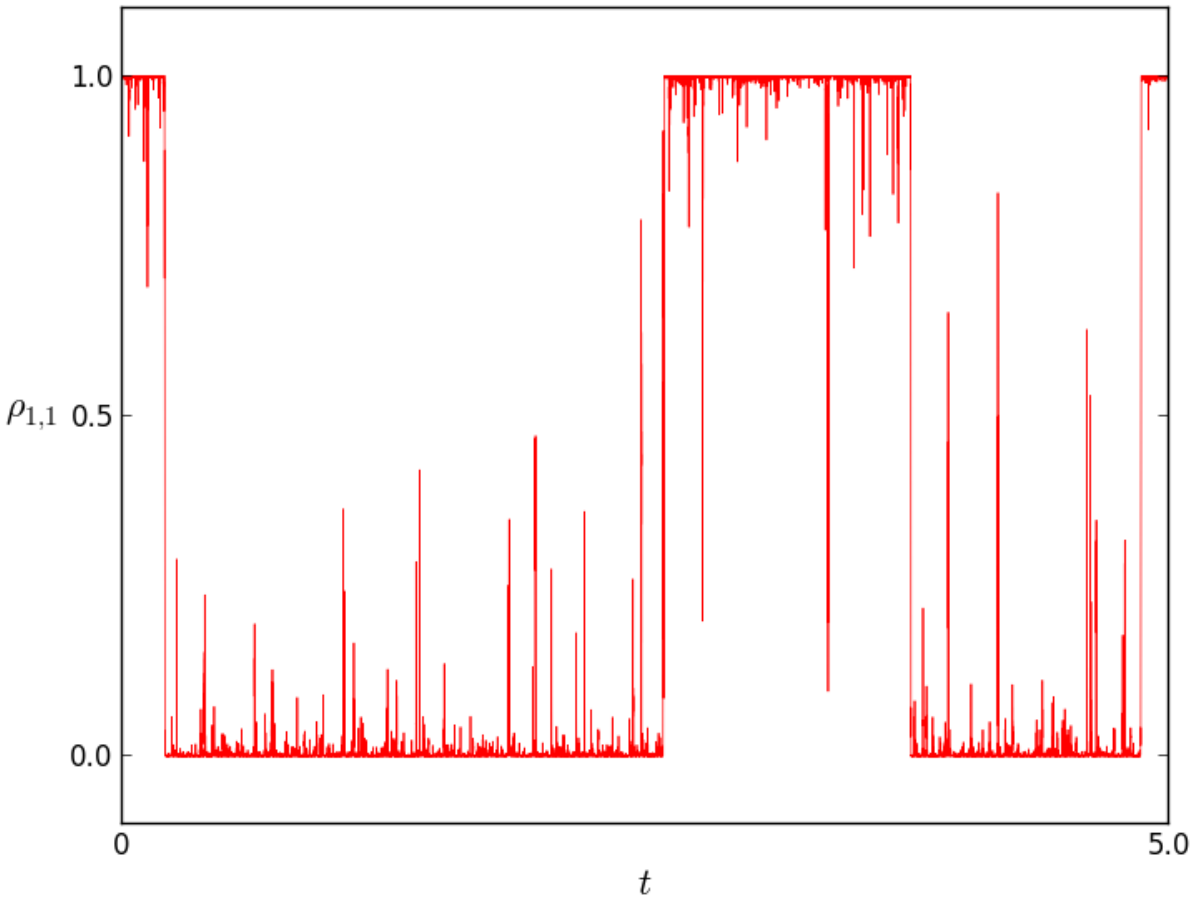
$\gamma = 5.0$

Results



$\gamma = 10$

Results



$\gamma = 20$

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With $\omega = \gamma\Omega$ (to avoid complete Zeno freezing), when $\gamma \to \infty$ and on a time interval without jumps, $\rho_{1,1}(t) = Q_t$ is a Poisson process of intensity:

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Remark

- The system density matrix stays pure during the spike (a probability spike is compensated by a phase spike)
- No simple epistemic interpretation this time!

DISCUSSION

Thank you for your time

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